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MAT 213 Brief Calculus

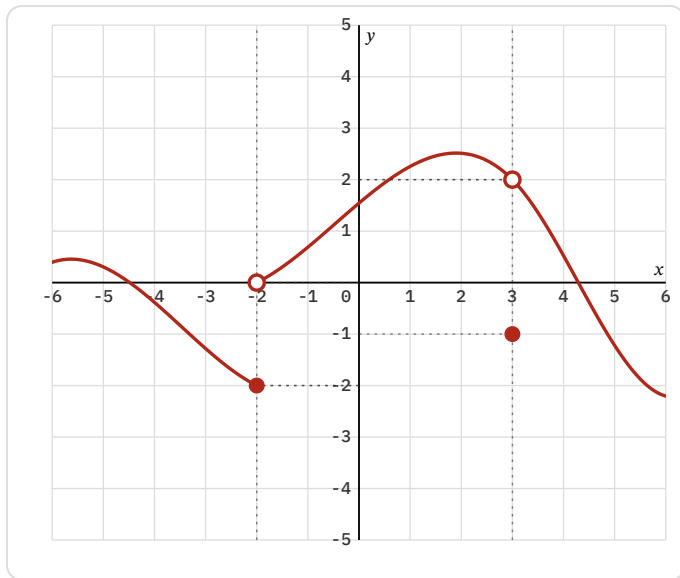
Mock Midterm B — Answer Key

Practice midterm (AI-Free) · Sections 1.1–1.4

Total: 58 points

1.

Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write DNE.



a.) $\lim_{x \rightarrow -2^-} f(x)$ (2)

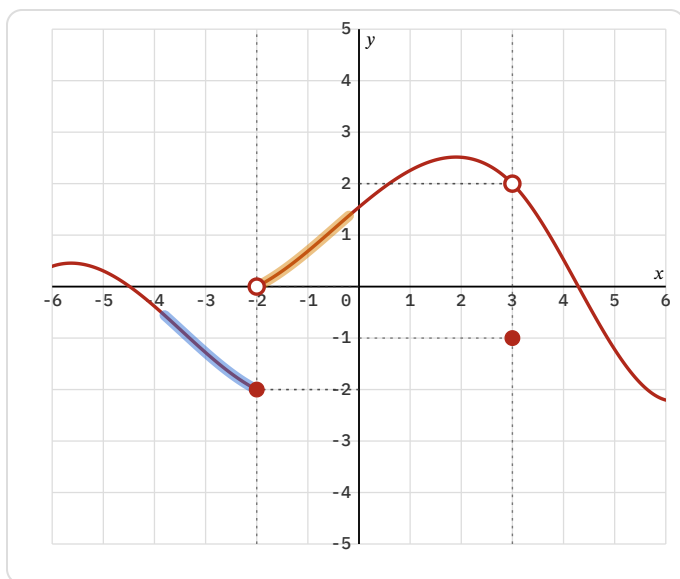
-2

b.) $\lim_{x \rightarrow 3^+} f(x)$ (2)

2

4 pts

WORKED SOLUTION



— approaching $x = -2$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

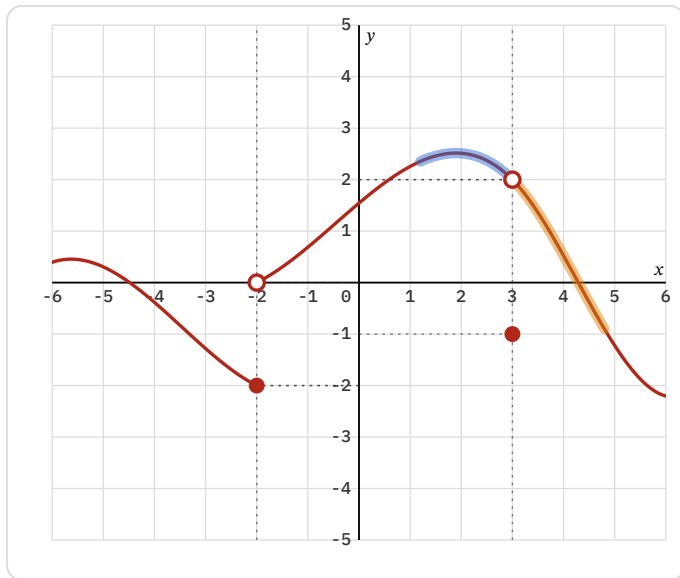
At $x = -2$

The **filled** dot at $x = -2$ sits at height -2, so $f(-2) = -2$.

Tracing the curve toward $x = -2$ from the left, the heights approach -2 (open or closed dot doesn't matter).

Tracing the curve toward $x = -2$ from the right, the heights approach 0 (open or closed dot doesn't matter).

The left and right limits don't match, so $\lim_{x \rightarrow -2} f(x)$ DNE.



— approaching $x = 3$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

At $x = 3$

The **filled** dot at $x = 3$ sits at height -1, so $f(3) = -1$.

Tracing the curve toward $x = 3$ from the left, the heights approach 2 (open or closed dot doesn't matter).

Tracing the curve toward $x = 3$ from the right, the heights approach 2 (open or closed dot doesn't matter).

Both one-sided limits equal 2, so $\lim_{x \rightarrow 3} f(x) = 2$ — even though $f(3) = -1$. The limit ignores the value AT the point.

2. Let $f(x) = \frac{x^3-27}{x-3}$. Use the table to see what the outputs do as x approaches **3**. 3 pts

x	2.99	2.999	2.9999	3	3.0001	3.001	3.01
$f(x)$	26.9101	26.991001	26.9991	?	27.0009	27.009001	27.0901

- a.) Can you find $\lim_{x \rightarrow 3} f(x)$ just by plugging in $x = 3$? (1)

(A) No — $x = 3$ makes a denominator 0, so $f(3)$ is undefined

- b.) $\lim_{x \rightarrow 3} f(x) =$ (2)

27

WORKED SOLUTION

Reading the left three columns, the outputs head toward **27**; the right three head toward **27**.

Both sides agree, so the limit is **27**.

$$\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3} = 27$$

Why: $\frac{(x-3)(x^2+3x+9)}{(x-3)} = x^2 + 3x + 9 \rightarrow 27$

3. Calculate each limit algebraically. 4 pts

a.) $\lim_{x \rightarrow -4} (2x^2 + 3x + 2)$ (2)

22

b.) $\lim_{x \rightarrow 2} \sqrt{x^2 + 21}$ (2)

5

WORKED SOLUTION

Each input is in the domain of its function, so direct substitution works.

a. $\lim_{x \rightarrow -4} (2x^2 + 3x + 2) = 2(-4)^2 + 3(-4) + 2 = 22$

b. $\lim_{x \rightarrow -3} \frac{x^2 - x - 9}{x - 2} = \frac{(-3)^2 - 1(-3) - 9}{-3 - 2} = \frac{3}{-5} = -\frac{3}{5}$

c. $\lim_{x \rightarrow 2} \sqrt{x^2 + 21} = \sqrt{(2)^2 + 21} = \sqrt{25} = 5$

4.

Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE.

a.) $\lim_{x \rightarrow 16} \frac{x-16}{\sqrt{x}-4}$ (2)
8

b.) $\lim_{h \rightarrow 0} \frac{(2+h)^3-8}{h}$ (2)
12

4 pts

WORKED SOLUTION

a.

$$\lim_{x \rightarrow 16} \frac{x-16}{\sqrt{x}-4} = \lim_{x \rightarrow 16} \frac{(\sqrt{x}-4)(\sqrt{x}+4)}{\sqrt{x}-4} = \lim_{x \rightarrow 16} (\sqrt{x}+4) = 8$$

b.

$$(2+h)^3 = 8 + 12h + 6h^2 + h^3, \text{ so } \lim_{h \rightarrow 0} \frac{(2+h)^3-8}{h} = \lim_{h \rightarrow 0} (12 + 6h + h^2) = 12$$

Notation tip: keep writing **lim** on every line until the step where you actually substitute.

5. Compute each limit algebraically. Use ∞ / $-\infty$ for unbounded results.

4 pts

a.) $\lim_{x \rightarrow -\infty} \frac{4x^2+1}{6x^2-9}$ (2)

$\frac{2}{3}$

b.) $\lim_{x \rightarrow \infty} \frac{x^5-7x^4+7x^3-3x^2+7x+4}{-4x^3+9x^2}$ (2)

$-\infty$

WORKED SOLUTION

a. Divide every term by x^2 , the highest power of x in the denominator:

$$\lim_{x \rightarrow -\infty} \frac{4x^2+1}{6x^2-9} = \lim_{x \rightarrow -\infty} \frac{4 + \frac{1}{x^2}}{6 - \frac{9}{x^2}}$$

Every fraction term $\rightarrow 0$, leaving $\frac{4}{6} = \frac{2}{3}$.

b. Divide every term by x^4 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{4}{-x^4 - 7x^2} = \lim_{x \rightarrow \infty} \frac{\frac{4}{x^4}}{-1 - \frac{7}{x^2}}$$

The top $\rightarrow 0$ while the bottom $\rightarrow -1$, so the limit is **0**.

c. Divide every term by x^3 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{x^5 - 7x^4 + 7x^3 - 3x^2 + 7x + 4}{-4x^3 + 9x^2} = \lim_{x \rightarrow \infty} \frac{x^2 - 7x + 7 - \frac{3}{x} + \frac{7}{x^2} + \frac{4}{x^3}}{-4 + \frac{9}{x}}$$

The bottom $\rightarrow -4$, but the top is dominated by $1x^2$, which $\rightarrow \infty$ as $x \rightarrow \infty$; dividing by -4 gives $-\infty$.

6.

Let

4 pts

$$G(x) = \begin{cases} -x^2 & x < -1 \\ 2x + 1 & x > -1 \end{cases}$$

Find each quantity. If it does not exist, write DNE.

a.) $G(-1)$ (2)

DNE

b.) $\lim_{x \rightarrow \infty} G(x)$ (2)

 ∞

WORKED SOLUTION

$G(-1)$: the pieces cover $x < -1$ and $x > -1$ only, so $G(-1)$ is undefined $\rightarrow$ DNE.

Left: use $-x^2$: $-(-1)^2 + 0 = -1$.

Right: use $2x + 1$: $2(-1) + 1 = -1$.

Two-sided: both sides give -1, so the limit is -1 (even though $G(-1)$ is undefined).

$G(2)$: $2 > -1$, so use $2x + 1$: $2(2) + 1 = 5$.

End behavior: as $x \rightarrow -\infty$ we're on the $-x^2$ piece — an even-degree term with negative leading coefficient $\rightarrow -\infty$. As $x \rightarrow \infty$ we're on the line with slope 2 $\rightarrow \infty$.

7. Let

4 pts

$$f(x) = \begin{cases} \frac{x^2 - 2x - 8}{x + 2} & x < -2 \\ 3x + 2 & x \geq -2 \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 0} f(x)$ (2)

2

b.) $\lim_{x \rightarrow -2} f(x)$ (2)

DNE

WORKED SOLUTION

For $x < -2$, factor the top: $\frac{(x+2)(x-4)}{x+2} = x - 4$ (valid since $x \neq -2$ there).

a. $0 > -2 \rightarrow$ bottom piece: $3(0) + 2 = 2$.

b. $-4 < -2 \rightarrow$ top piece, simplified: $-4 - 4 = -8$.

c. The bottom piece includes $x = -2$ ($\geq$), so $f(-2) = 3(-2) + 2 = -4$.

d. Left: $\lim_{x \rightarrow -2^-} (x - 4) = -6$.

e. Right: $\lim_{x \rightarrow -2^+} (3x + 2) = -4$.

f. $-6 \neq -4$, so the limit DNE.

8. Find the value of k that makes f continuous at $x = -2$.

4 pts

$$f(x) = \begin{cases} x^2 + x & x < -2 \\ kx + 6 & x \geq -2 \end{cases}$$

a.) $\lim_{x \rightarrow -2^-} f(x)$ (2)

2

b.) $k =$ (2)

2

WORKED SOLUTION

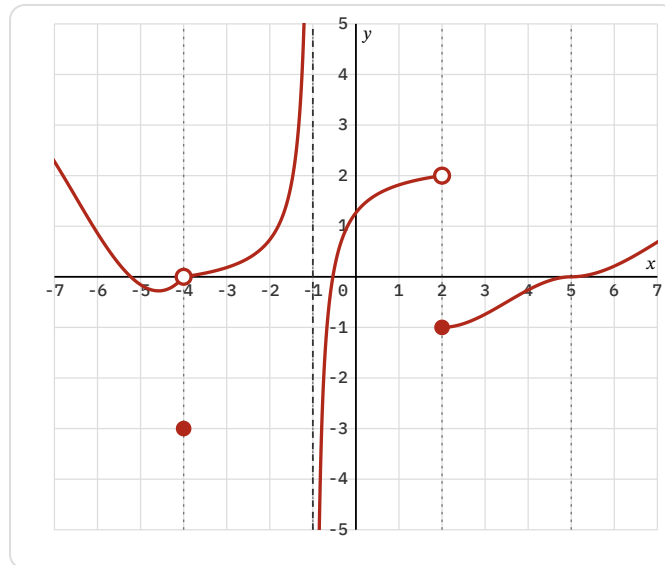
Left: $(-2)^2 + 1(-2) = 2$. Right (and $f(-2)$): $k(-2) + 6$.

$$-2k + 6 = 2 \Rightarrow k = \frac{-4}{-2} = 2$$

9.

Classify the behavior of f at each labeled x -value.

2 pts

a.) At $x = -4$ (1)

(B) Removable discontinuity

b.) At $x = 5$ (1)

(A) Continuous

WORKED SOLUTION

$x = -4$: Both sides approach 0, but the filled dot is at -3 $\rightarrow$ removable (the limit exists, it just doesn't match f).

$x = -1$: The graph shoots off along a vertical asymptote $\rightarrow$ infinite.

$x = 2$: Left approaches 2, right approaches -1 $\rightarrow$ jump.

$x = 5$: No break, and the dot is on the curve $\rightarrow$ continuous.

10.

Let $V(t) = t^2 - 50t + 300$, where $V(t)$ represents the number of gallons of water in a tank t minutes after a drain is opened. Find the average rate of change over $[6, 8]$.

a.) $V(6)$ (2)

36

b.) Which sentence correctly states the result **with units**? (1)

(C) Between $t = 6$ and $t = 8$ minutes, the amount of water in the tank changes at an average rate of -36 gallons per minute.

3 pts

WORKED SOLUTION

$$V(6) = 36, \quad V(8) = -36$$

$$\frac{V(8) - V(6)}{8 - 6} = \frac{-36 - (36)}{2} = \frac{-72}{2} = -36 \text{ gallons per minute}$$

It's an *average* over an interval (not an instant), and the units are gallons per minute — gallons per minute. The total change would be -72 gallons.

11.

Let $f(x) = \frac{-2}{x}$. Calculate the average rate of change of f over each interval. (Rounded answers to 2+ decimal places are fine.)

a.) $[3, 5]$ (2)

$$\frac{2}{15}$$

b.) As the interval shrinks toward $x = 3$, these values approach... (1)

$$(A) \frac{2}{9}$$

3 pts

WORKED SOLUTION

$$[3, 5] : \frac{f(5) - f(3)}{2} = \frac{2}{15} \approx 0.1333$$

$$[3, 4] : \frac{f(4) - f(3)}{1} = \frac{1}{6} \approx 0.1667$$

$$[3, 3.1] : \frac{f(3.1) - f(3)}{0.1} = \frac{20}{93} \approx 0.2151$$

$$[3, 3.01] : \frac{f(3.01) - f(3)}{0.01} = \frac{200}{903} \approx 0.2215$$

Each value is the slope of a secant line through $(3, f(3))$. As the interval shrinks they approach the slope of the **tangent line**: $f'(x) = -\frac{2}{x^2}$, so $f'(3) = \frac{2}{9}$.

12.

Let $f(x) = x^3 - 3x$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table.

a.) Simplified difference quotient (in terms of x and h) (3)

$$3x^2 + 3xh + h^2 - 3$$

b.) As $h \rightarrow 0$, what value do these approach? (2)

$$0$$

5 pts

WORKED SOLUTION

$$f(x+h) = (x+h)^3 - 3(x+h) = x^3 + 3x^2h + 3xh^2 + h^3 - 3x - 3h$$

$$\frac{f(x+h) - f(x)}{h} = \frac{3x^2h + 3xh^2 + h^3 - 3h}{h} = 3x^2 + 3xh + h^2 - 3$$

Now substitute:

$$x = -1, h = 1 : \quad -2$$

$$x = -1, h = 0.1 : \quad -\frac{29}{100} = -0.29$$

$$x = -1, h = 0.01 : \quad -\frac{299}{10000} = -0.0299$$

Letting $h \rightarrow 0$ leaves 0: the instantaneous rate of change (slope of the tangent line) at $x = -1$. That's exactly what section 1.4 formalizes.

13.

Let $f(x) = 3x^2 - 2x - 6$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$.

a.) $f'(x) =$ (3)
 $6x - 2$

b.) What does $f'(-3) = -20$ mean? (1)
(D) The instantaneous rate of change of f at $x = -3$ is -20 — the slope of the tangent line there.

4 pts

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[3(x+h)^2 - 2(x+h) - 6] - (3x^2 - 2x - 6)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 - 2x - 2h - 6 - 3x^2 + 2x + 6}{h} \\ &= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 2h}{h} = \lim_{h \rightarrow 0} \frac{h(6x + 3h - 2)}{h} \\ &= \lim_{h \rightarrow 0} (6x + 3h - 2) = 6x - 2 \end{aligned}$$

Then $f'(-3) = -20$ and the slope of the tangent at $x = 2$ is $f'(2) = 10$.

$f'(-3)$ is the *instantaneous* rate of change at one input — the slope of the tangent line, not the height of the graph and not an average over an interval.

14. Let $f(x) = x^2 - 8x - 3$, so $f'(x) = 2x - 8$. 4 pts

a.) At what x is the tangent line horizontal? (2)

4

b.) At what x does the tangent line have slope 2? (2)

5

WORKED SOLUTION

$$f'(x) = 0 : \quad 2x - 8 = 0 \Rightarrow x = 4$$

$$f(4) = -19$$

$$f'(x) = 2 : \quad 2x - 8 = 2 \Rightarrow x = \frac{10}{2} = 5$$

15.

Let $h(t) = -16t^2 + 128t$ be the height, in feet, of a rocket t seconds after launch.

a.) $h'(t) =$ (3)
 $-32t + 128$

b.) Interpret $h'(5) = -32$. (1)

(C) At $t = 5$ seconds, the height of the rocket is decreasing at a rate of 32 feet per second.

4 pts

WORKED SOLUTION

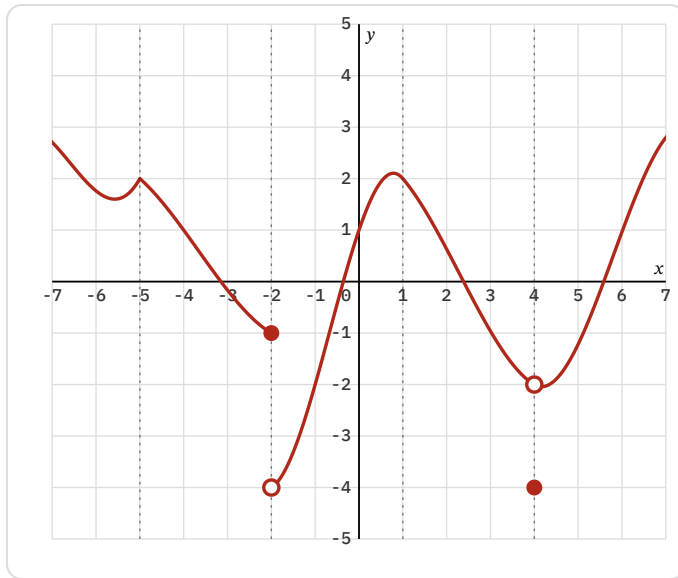
$$\begin{aligned} h'(t) &= \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} = \lim_{h \rightarrow 0} \frac{[-16(t+h)^2 + 128(t+h)] - (-16t^2 + 128t)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-16t^2 - 32th - 16h^2 + 128t + 128h + 16t^2 - 128t}{h} \\ &= \lim_{h \rightarrow 0} \frac{-32th - 16h^2 + 128h}{h} = \lim_{h \rightarrow 0} \frac{h(-32t - 16h + 128)}{h} \\ &= \lim_{h \rightarrow 0} (-32t - 16h + 128) = -32t + 128 \end{aligned}$$

$$h'(5) = -32(5) + 128 = -32$$

Positive/negative tells the direction, and the units are feet per second: at that instant the height of the rocket is decreasing at 32 feet per second. (The value $h(5) = 240$ is a different quantity.)

16.

At each labeled x -value, decide whether the graph of f is continuous there and whether it is differentiable there.



a.) At $x = -5$ (1)

(B) Continuous but NOT differentiable

b.) At $x = 4$ (1)

(C) Not continuous (so not differentiable)

2 pts

WORKED SOLUTION

$x = -5$: No break, but the curve comes to a sharp point (corner): the slopes from the left and right disagree $\rightarrow$ continuous, not differentiable.

$x = -2$: The graph jumps $\rightarrow$ not continuous, so not differentiable.

$x = 1$: Smooth and unbroken — a single tangent line fits $\rightarrow$ continuous and differentiable.

$x = 4$: There's a hole (the dot isn't on the curve) $\rightarrow$ not continuous, so not differentiable.
